

# System size and energy dependence of near-side di-hadron correlations

H. Agakishiev,<sup>18</sup> M. M. Aggarwal,<sup>30</sup> Z. Ahammed,<sup>22</sup> A. V. Alakhverdyants,<sup>18</sup> I. Alekseev,<sup>16</sup> J. Alford,<sup>19</sup>  
 B. D. Anderson,<sup>19</sup> C. D. Anson,<sup>28</sup> D. Arkhipkin,<sup>3</sup> G. S. Averichev,<sup>18</sup> J. Balewski,<sup>23</sup> L. S. Barnby,<sup>2</sup> D. R. Beavis,<sup>3</sup>  
 N. K. Behera,<sup>14</sup> R. Bellwied,<sup>44</sup> M. J. Betancourt,<sup>23</sup> R. R. Betts,<sup>8</sup> A. Bhasin,<sup>17</sup> A. K. Bhati,<sup>30</sup> H. Bichsel,<sup>51</sup>  
 J. Bielcik,<sup>10</sup> J. Bielcikova,<sup>11</sup> B. Biritz,<sup>6</sup> L. C. Bland,<sup>3</sup> M. Bombara,<sup>2</sup> I. G. Bordyuzhin,<sup>16</sup> W. Borowski,<sup>41</sup>  
 J. Bouchet,<sup>19</sup> E. Braidot,<sup>27</sup> A. V. Brandin,<sup>26</sup> A. Bridgeman,<sup>1</sup> S. G. Brovko,<sup>5</sup> E. Bruna,<sup>54</sup> S. Bueltmann,<sup>29</sup>  
 I. Bunzarov,<sup>18</sup> T. P. Burton,<sup>3</sup> X. Z. Cai,<sup>40</sup> H. Caines,<sup>54</sup> M. Calderón de la Barca Sánchez,<sup>5</sup> D. Cebra,<sup>5</sup> R. Cendejas,<sup>6</sup>  
 M. C. Cervantes,<sup>42</sup> Z. Chajecki,<sup>28</sup> P. Chaloupka,<sup>11</sup> S. Chattopadhyay,<sup>49</sup> H. F. Chen,<sup>38</sup> J. H. Chen,<sup>40</sup>  
 J. Y. Chen,<sup>53</sup> L. Chen,<sup>53</sup> J. Cheng,<sup>46</sup> M. Cherney,<sup>9</sup> A. Chikanian,<sup>54</sup> K. E. Choi,<sup>34</sup> W. Christie,<sup>3</sup> P. Chung,<sup>11</sup>  
 M. J. M. Codrington,<sup>42</sup> R. Corliss,<sup>23</sup> J. G. Cramer,<sup>51</sup> H. J. Crawford,<sup>4</sup> S. Dash,<sup>13</sup> A. Davila Leyva,<sup>43</sup>  
 L. C. De Silva,<sup>44</sup> R. R. Debbé,<sup>3</sup> T. G. Dedovich,<sup>18</sup> A. A. Derevschikov,<sup>32</sup> R. Derradi de Souza,<sup>7</sup> L. Didenko,<sup>3</sup>  
 P. Djawotho,<sup>42</sup> S. M. Dogra,<sup>17</sup> X. Dong,<sup>22</sup> J. L. Drachenberg,<sup>42</sup> J. E. Draper,<sup>5</sup> J. C. Dunlop,<sup>3</sup> L. G. Efimov,<sup>18</sup>  
 M. Elnimr,<sup>52</sup> J. Engelage,<sup>4</sup> G. Eppley,<sup>36</sup> M. Estienne,<sup>41</sup> L. Eun,<sup>31</sup> O. Evdokimov,<sup>8</sup> R. Fatemi,<sup>20</sup> J. Fedorisin,<sup>18</sup>  
 R. G. Fersch,<sup>20</sup> P. Filip,<sup>18</sup> E. Finch,<sup>54</sup> V. Fine,<sup>3</sup> Y. Fisyak,<sup>3</sup> C. A. Gagliardi,<sup>42</sup> L. Gaillard,<sup>2</sup> D. R. Gangadharan,<sup>6</sup>  
 F. Geurts,<sup>36</sup> P. Ghosh,<sup>49</sup> Y. N. Gorbunov,<sup>9</sup> A. Gordon,<sup>3</sup> O. G. Grebenyuk,<sup>22</sup> D. Grosnick,<sup>48</sup> S. M. Guertin,<sup>6</sup>  
 A. Gupta,<sup>17</sup> W. Guryan,<sup>3</sup> B. Haag,<sup>5</sup> O. Hajkova,<sup>10</sup> A. Hamed,<sup>42</sup> L-X. Han,<sup>40</sup> J. W. Harris,<sup>54</sup> J. P. Hays-Wehle,<sup>23</sup>  
 M. Heinz,<sup>54</sup> S. Heppelmann,<sup>31</sup> A. Hirsch,<sup>33</sup> E. Hjort,<sup>22</sup> G. W. Hoffmann,<sup>43</sup> D. J. Hofman,<sup>8</sup> B. Huang,<sup>38</sup>  
 H. Z. Huang,<sup>6</sup> T. J. Humanic,<sup>28</sup> L. Huo,<sup>42</sup> G. Igo,<sup>6</sup> P. Jacobs,<sup>22</sup> W. W. Jacobs,<sup>15</sup> C. Jena,<sup>13</sup> F. Jin,<sup>40</sup> P. G. Jones,<sup>2</sup>  
 J. Joseph,<sup>19</sup> E. G. Judd,<sup>4</sup> S. Kabana,<sup>41</sup> K. Kang,<sup>46</sup> J. Kapitan,<sup>11</sup> K. Kauder,<sup>8</sup> H. W. Ke,<sup>53</sup> D. Keane,<sup>19</sup>  
 A. Kechechyan,<sup>18</sup> D. Kettler,<sup>51</sup> D. P. Kikola,<sup>33</sup> J. Kiryluk,<sup>22</sup> A. Kisiel,<sup>50</sup> V. Kizka,<sup>18</sup> S. R. Klein,<sup>22</sup> A. G. Knospe,<sup>54</sup>  
 D. D. Koetke,<sup>48</sup> T. Kollegger,<sup>12</sup> J. Konzer,<sup>33</sup> I. Koralt,<sup>29</sup> L. Koroleva,<sup>16</sup> W. Korsch,<sup>20</sup> L. Kotchenda,<sup>26</sup>  
 V. Kouchpil,<sup>11</sup> P. Kravtsov,<sup>26</sup> K. Krueger,<sup>1</sup> M. Krus,<sup>10</sup> L. Kumar,<sup>19</sup> P. Kurnadi,<sup>6</sup> M. A. C. Lamont,<sup>3</sup>  
 J. M. Landgraf,<sup>3</sup> S. LaPointe,<sup>52</sup> J. Lauret,<sup>3</sup> A. Lebedev,<sup>3</sup> R. Lednický,<sup>18</sup> J. H. Lee,<sup>3</sup> W. Leight,<sup>23</sup> M. J. LeVine,<sup>3</sup>  
 C. Li,<sup>38</sup> L. Li,<sup>43</sup> N. Li,<sup>53</sup> W. Li,<sup>40</sup> X. Li,<sup>33</sup> X. Li,<sup>39</sup> Y. Li,<sup>46</sup> Z. M. Li,<sup>53</sup> M. A. Lisa,<sup>28</sup> F. Liu,<sup>53</sup> H. Liu,<sup>5</sup> J. Liu,<sup>36</sup>  
 T. Ljubicic,<sup>3</sup> W. J. Llope,<sup>36</sup> R. S. Longacre,<sup>3</sup> W. A. Love,<sup>3</sup> Y. Lu,<sup>38</sup> E. V. Lukashov,<sup>26</sup> X. Luo,<sup>38</sup> G. L. Ma,<sup>40</sup>  
 Y. G. Ma,<sup>40</sup> D. P. Mahapatra,<sup>13</sup> R. Majka,<sup>54</sup> O. I. Mall,<sup>5</sup> L. K. Mangotra,<sup>17</sup> R. Manweiler,<sup>48</sup> S. Margetis,<sup>19</sup>  
 C. Markert,<sup>43</sup> H. Masui,<sup>22</sup> H. S. Matis,<sup>22</sup> Yu. A. Matulenko,<sup>32</sup> D. McDonald,<sup>36</sup> T. S. McShane,<sup>9</sup> A. Meschanin,<sup>32</sup>  
 R. Milner,<sup>23</sup> N. G. Minaev,<sup>32</sup> S. Mioduszewski,<sup>42</sup> M. K. Mitrovski,<sup>3</sup> Y. Mohammed,<sup>42</sup> B. Mohanty,<sup>49</sup>  
 M. M. Mondal,<sup>49</sup> B. Morozov,<sup>16</sup> D. A. Morozov,<sup>32</sup> M. G. Munhoz,<sup>37</sup> M. K. Mustafa,<sup>33</sup> M. Naglis,<sup>22</sup> B. K. Nandi,<sup>14</sup>  
 C. Nattrass,<sup>54</sup> T. K. Nayak,<sup>49</sup> J. M. Nelson,<sup>2</sup> P. K. Netrakanti,<sup>33</sup> L. V. Nogach,<sup>32</sup> S. B. Nurushev,<sup>32</sup> G. Odyniec,<sup>22</sup>  
 A. Ogawa,<sup>3</sup> K. Oh,<sup>34</sup> A. Ohlson,<sup>54</sup> V. Okorokov,<sup>26</sup> E. W. Oldag,<sup>43</sup> D. Olson,<sup>22</sup> M. Pachr,<sup>10</sup> B. S. Page,<sup>15</sup>  
 S. K. Pal,<sup>49</sup> Y. Pandit,<sup>19</sup> Y. Panebratsev,<sup>18</sup> T. Pawlak,<sup>50</sup> H. Pei,<sup>8</sup> T. Peitzmann,<sup>27</sup> C. Perkins,<sup>4</sup> W. Peryt,<sup>50</sup>  
 S. C. Phatak,<sup>13</sup> P. Pile,<sup>3</sup> M. Planinic,<sup>55</sup> M. A. Ploskon,<sup>22</sup> J. Pluta,<sup>50</sup> D. Plyku,<sup>29</sup> N. Poljak,<sup>55</sup> J. Porter,<sup>22</sup>  
 A. M. Poskanzer,<sup>22</sup> B. V. K. S. Potukuchi,<sup>17</sup> C. B. Powell,<sup>22</sup> D. Prindle,<sup>51</sup> C. Pruneau,<sup>52</sup> N. K. Pruthi,<sup>30</sup>  
 P. R. Pujahari,<sup>14</sup> J. Putschke,<sup>54</sup> H. Qiu,<sup>21</sup> R. Raniwala,<sup>35</sup> S. Raniwala,<sup>35</sup> R. L. Ray,<sup>43</sup> R. Redwine,<sup>23</sup> R. Reed,<sup>5</sup>  
 H. G. Ritter,<sup>22</sup> J. B. Roberts,<sup>36</sup> O. V. Rogachevskiy,<sup>18</sup> J. L. Romero,<sup>5</sup> L. Ruan,<sup>3</sup> J. Rusnak,<sup>11</sup> N. R. Sahoo,<sup>49</sup>  
 I. Sakrejda,<sup>22</sup> S. Salur,<sup>5</sup> J. Sandweiss,<sup>54</sup> E. Sangaline,<sup>5</sup> A. Sarkar,<sup>14</sup> J. Schambach,<sup>43</sup> R. P. Scharenberg,<sup>33</sup>  
 A. M. Schmah,<sup>22</sup> N. Schmitz,<sup>24</sup> T. R. Schuster,<sup>12</sup> J. Seele,<sup>23</sup> J. Seger,<sup>9</sup> I. Selyuzhenkov,<sup>15</sup> P. Seyboth,<sup>24</sup>  
 E. Shalahiev,<sup>18</sup> M. Shao,<sup>38</sup> M. Sharma,<sup>52</sup> S. S. Shi,<sup>53</sup> Q. Y. Shou,<sup>40</sup> E. P. Sichtermann,<sup>22</sup> F. Simon,<sup>24</sup>  
 R. N. Singaraju,<sup>49</sup> M. J. Skoby,<sup>33</sup> N. Smirnov,<sup>54</sup> P. Sorensen,<sup>3</sup> H. M. Spinka,<sup>1</sup> B. Srivastava,<sup>33</sup> T. D. S. Stanislaus,<sup>48</sup>  
 D. Staszak,<sup>6</sup> S. G. Steadman,<sup>23</sup> J. R. Stevens,<sup>15</sup> R. Stock,<sup>12</sup> M. Strikhanov,<sup>26</sup> B. Stringfellow,<sup>33</sup> A. A. P. Suaide,<sup>37</sup>  
 M. C. Suarez,<sup>8</sup> N. L. Subba,<sup>19</sup> M. Sumner,<sup>11</sup> X. M. Sun,<sup>22</sup> Y. Sun,<sup>38</sup> Z. Sun,<sup>21</sup> B. Surrow,<sup>23</sup> D. N. Svirida,<sup>16</sup>  
 T. J. M. Symons,<sup>22</sup> A. Szanto de Toledo,<sup>37</sup> J. Takahashi,<sup>7</sup> A. H. Tang,<sup>3</sup> Z. Tang,<sup>38</sup> L. H. Tarini,<sup>52</sup> T. Tarnowsky,<sup>25</sup>  
 D. Thein,<sup>43</sup> J. H. Thomas,<sup>22</sup> J. Tian,<sup>40</sup> A. R. Timmins,<sup>44</sup> D. Tlusty,<sup>11</sup> M. Tokarev,<sup>18</sup> T. A. Trainor,<sup>51</sup>  
 S. Trentalange,<sup>6</sup> R. E. Tribble,<sup>42</sup> P. Tribedy,<sup>49</sup> O. D. Tsai,<sup>6</sup> T. Ullrich,<sup>3</sup> D. G. Underwood,<sup>1</sup> G. Van Buren,<sup>3</sup>  
 G. van Nieuwenhuizen,<sup>23</sup> J. A. Vanfossen, Jr.,<sup>19</sup> R. Varma,<sup>14</sup> G. M. S. Vasconcelos,<sup>7</sup> A. N. Vasiliev,<sup>32</sup> F. Videbæk,<sup>3</sup>  
 Y. P. Viyogi,<sup>49</sup> S. Vokal,<sup>18</sup> S. A. Voloshin,<sup>52</sup> M. Wada,<sup>43</sup> M. Walker,<sup>23</sup> F. Wang,<sup>33</sup> G. Wang,<sup>6</sup> H. Wang,<sup>25</sup>  
 J. S. Wang,<sup>21</sup> Q. Wang,<sup>33</sup> X. L. Wang,<sup>38</sup> Y. Wang,<sup>46</sup> G. Webb,<sup>20</sup> J. C. Webb,<sup>3</sup> G. D. Westfall,<sup>25</sup> C. Whitten Jr.,<sup>6</sup>  
 H. Wieman,<sup>22</sup> S. W. Wissink,<sup>15</sup> R. Witt,<sup>47</sup> W. Witzke,<sup>20</sup> Y. F. Wu,<sup>53</sup> Z. Xiao,<sup>46</sup> W. Xie,<sup>33</sup> H. Xu,<sup>21</sup> N. Xu,<sup>22</sup>  
 Q. H. Xu,<sup>39</sup> W. Xu,<sup>6</sup> Y. Xu,<sup>38</sup> Z. Xu,<sup>3</sup> L. Xue,<sup>40</sup> Y. Yang,<sup>21</sup> Y. Yang,<sup>53</sup> P. Yepes,<sup>36</sup> K. Yip,<sup>3</sup> I-K. Yoo,<sup>34</sup>  
 M. Zawisza,<sup>50</sup> H. Zbroszczyk,<sup>50</sup> W. Zhan,<sup>21</sup> J. B. Zhang,<sup>53</sup> S. Zhang,<sup>40</sup> W. M. Zhang,<sup>19</sup> X. P. Zhang,<sup>46</sup> Y. Zhang,<sup>22</sup>  
 Z. P. Zhang,<sup>38</sup> J. Zhao,<sup>40</sup> C. Zhong,<sup>40</sup> W. Zhou,<sup>39</sup> X. Zhu,<sup>46</sup> Y. H. Zhu,<sup>40</sup> R. Zoulikarneeve,<sup>18</sup> and Y. Zoulikarneeve<sup>18</sup>

## (STAR Collaboration)

- <sup>1</sup>Argonne National Laboratory, Argonne, Illinois 60439, USA  
<sup>2</sup>University of Birmingham, Birmingham, United Kingdom  
<sup>3</sup>Brookhaven National Laboratory, Upton, New York 11973, USA  
<sup>4</sup>University of California, Berkeley, California 94720, USA  
<sup>5</sup>University of California, Davis, California 95616, USA  
<sup>6</sup>University of California, Los Angeles, California 90095, USA  
<sup>7</sup>Universidade Estadual de Campinas, Sao Paulo, Brazil  
<sup>8</sup>University of Illinois at Chicago, Chicago, Illinois 60607, USA  
<sup>9</sup>Creighton University, Omaha, Nebraska 68178, USA  
<sup>10</sup>Czech Technical University in Prague, FNSPE, Prague, 115 19, Czech Republic  
<sup>11</sup>Nuclear Physics Institute AS CR, 250 68 Řež/Prague, Czech Republic  
<sup>12</sup>University of Frankfurt, Frankfurt, Germany  
<sup>13</sup>Institute of Physics, Bhubaneswar 751005, India  
<sup>14</sup>Indian Institute of Technology, Mumbai, India  
<sup>15</sup>Indiana University, Bloomington, Indiana 47408, USA  
<sup>16</sup>Alikhanov Institute for Theoretical and Experimental Physics, Moscow, Russia  
<sup>17</sup>University of Jammu, Jammu 180001, India  
<sup>18</sup>Joint Institute for Nuclear Research, Dubna, 141 980, Russia  
<sup>19</sup>Kent State University, Kent, Ohio 44242, USA  
<sup>20</sup>University of Kentucky, Lexington, Kentucky, 40506-0055, USA  
<sup>21</sup>Institute of Modern Physics, Lanzhou, China  
<sup>22</sup>Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA  
<sup>23</sup>Massachusetts Institute of Technology, Cambridge, MA 02139-4307, USA  
<sup>24</sup>Max-Planck-Institut für Physik, Munich, Germany  
<sup>25</sup>Michigan State University, East Lansing, Michigan 48824, USA  
<sup>26</sup>Moscow Engineering Physics Institute, Moscow Russia  
<sup>27</sup>NIKHEF and Utrecht University, Amsterdam, The Netherlands  
<sup>28</sup>Ohio State University, Columbus, Ohio 43210, USA  
<sup>29</sup>Old Dominion University, Norfolk, VA, 23529, USA  
<sup>30</sup>Panjab University, Chandigarh 160014, India  
<sup>31</sup>Pennsylvania State University, University Park, Pennsylvania 16802, USA  
<sup>32</sup>Institute of High Energy Physics, Protvino, Russia  
<sup>33</sup>Purdue University, West Lafayette, Indiana 47907, USA  
<sup>34</sup>Pusan National University, Pusan, Republic of Korea  
<sup>35</sup>University of Rajasthan, Jaipur 302004, India  
<sup>36</sup>Rice University, Houston, Texas 77251, USA  
<sup>37</sup>Universidade de Sao Paulo, Sao Paulo, Brazil  
<sup>38</sup>University of Science & Technology of China, Hefei 230026, China  
<sup>39</sup>Shandong University, Jinan, Shandong 250100, China  
<sup>40</sup>Shanghai Institute of Applied Physics, Shanghai 201800, China  
<sup>41</sup>SUBATECH, Nantes, France  
<sup>42</sup>Texas A&M University, College Station, Texas 77843, USA  
<sup>43</sup>University of Texas, Austin, Texas 78712, USA  
<sup>44</sup>University of Houston, Houston, TX, 77204, USA  
<sup>45</sup>University of Tennessee, Knoxville, TN, 37996, USA  
<sup>46</sup>Tsinghua University, Beijing 100084, China  
<sup>47</sup>United States Naval Academy, Annapolis, MD 21402, USA  
<sup>48</sup>Valparaiso University, Valparaiso, Indiana 46383, USA  
<sup>49</sup>Variable Energy Cyclotron Centre, Kolkata 700064, India  
<sup>50</sup>Warsaw University of Technology, Warsaw, Poland  
<sup>51</sup>University of Washington, Seattle, Washington 98195, USA  
<sup>52</sup>Wayne State University, Detroit, Michigan 48201, USA  
<sup>53</sup>Institute of Particle Physics, CCNU (HZNU), Wuhan 430079, China  
<sup>54</sup>Yale University, New Haven, Connecticut 06520, USA  
<sup>55</sup>University of Zagreb, Zagreb, HR-10002, Croatia

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Two-particle azimuthal ( $\Delta\phi$ ) and pseudorapidity ( $\Delta\eta$ ) correlations using a trigger particle with large transverse momentum ( $p_T$ ) in d+Au, Cu+Cu and Au+Au collisions at  $\sqrt{s_{NN}} = 62.4$  GeV and 200 GeV from the STAR experiment at RHIC are presented. The near-side correlation is separated into a jet-like component, narrow in both  $\Delta\phi$  and  $\Delta\eta$ , and the ridge, narrow in  $\Delta\phi$  but broad in  $\Delta\eta$ . Both components are studied as a function of collision centrality, and the jet-like correlation is studied as a function of the trigger and associated  $p_T$ . The behavior of the jet-like component

is remarkably consistent for different collision systems, suggesting it is produced by fragmentation. The width of the jet-like correlation is found to increase with the system size. The ridge, previously observed in Au+Au collisions at  $\sqrt{s_{NN}} = 200$  GeV, is also found in Cu+Cu collisions and in collisions at  $\sqrt{s_{NN}} = 62.4$  GeV, but is found to be substantially smaller at  $\sqrt{s_{NN}} = 62.4$  GeV than at  $\sqrt{s_{NN}} = 200$  GeV for the same number of participants ( $N_{part}$ ). Measurements of the ridge are compared to models.

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## I. INTRODUCTION

Jets are a useful probe of the hot, dense medium created in heavy-ion collisions at the Relativistic Heavy Ion Collider (RHIC) at Brookhaven National Laboratory. Jet quenching [1] was first observed as the suppression of inclusive hadron spectra at large transverse momenta ( $p_T$ ) in central Au+Au collisions with respect to p+p scaled by number of binary nucleon-nucleon collisions [2–7]. Properties of jets at RHIC have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum ( $p_T$ ) [8–14].

Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed their significant modification in Au+Au relative to p+p and d+Au collisions at the top RHIC energy of  $\sqrt{s_{NN}} = 200$  GeV. For low  $p_T^{associated}$ , the amplitude of the away-side peak is greater and the shape is modified in Au+Au collisions [8, 9]. At intermediate  $p_T$  ( $3 < p_T^{trigger} < 6$  GeV/c,  $2 < p_T^{associated} < 4$  GeV/c), the away-side correlation peak is strongly suppressed [10]. At higher  $p_T$ , the away-side peak reappears without shape modification, but the away-side per trigger yield is smaller in Au+Au collisions than in p+p and d+Au [11].

The associated particle distribution on the near side of the trigger particle, the subject of this paper, is also significantly modified in central Au+Au collisions. In p+p and d+Au collisions, there is a peak narrow in azimuth ( $\Delta\phi$ ) and pseudorapidity ( $\Delta\eta$ ) around the trigger particle, which we refer to as the jet-like correlation. This peak is also present in Au+Au collisions, but an additional structure which is narrow in azimuth but broad in pseudorapidity has been observed in central Au+Au collisions at  $\sqrt{s_{NN}} = 200$  GeV [8, 12–14]. This structure, called the ridge, is independent of  $\Delta\eta$  within the STAR acceptance,  $|\Delta\eta| < 2.0$ , within errors, and persists to high  $p_T$  ( $p_T^{trigger} \approx 6$  GeV/c,  $p_T^{associated} \approx 3$  GeV/c). While the spectrum of particles in the jet-like correlation becomes flatter with increasing  $p_T^{trigger}$ , the slope of the spectrum of particles in the ridge is independent of  $p_T^{trigger}$  and closer to the inclusive spectrum than to that of the jet-like correlation. Recent studies by PHOBOS of di-hadron correlations at lower transverse momenta ( $p_T^{trigger} > 2.5$  GeV/c,  $p_T^{associated} > 20$  MeV/c) show that the ridge is roughly independent of  $\Delta\eta$  and extends over four units in  $\Delta\eta$  [15]. A similar broad correlation in pseudorapidity is also evident in untriggered di-hadron correlations [16].

Several mechanisms for the production of the ridge

have been proposed since the first observation of this new phenomenon. In one model [17] the ridge is proposed to be formed from gluon radiation emitted by a high- $p_T$  parton propagating in the medium with strong longitudinal flow. The momentum-kick model proposes that the ridge forms as a fast parton traveling through the medium loses energy through collisions with partons in the medium, causing those partons to be correlated in space with the fast parton [18–20]. Parton recombination has been also proposed as a mechanism for the production of the ridge [21–23]. Another model [24, 25] suggest that the ridge is not actually caused by a hard parton but is the product of radial flow and the surface biased emission of the trigger particle, causing an apparent correlation between particles from the bulk and high- $p_T$  trigger particles.

Another large class of models is based on the conversion of correlations in the initial state into momentum space through various flow effects. The model in [26] explains the ridge as arising from the spontaneous formation of extended color fields in a longitudinally expanding medium due to the presence of plasma instabilities. Long-range pseudorapidity correlations formed in an initial state glasma combined with transverse radial flow have been also discussed as a mechanism for the ridge [27–29]. Recently, it has been suggested that triangular anisotropy in the initial collision geometry caused by event-by-event fluctuations can give rise to triangular flow, which leads to the ridge and contributes to the double peaked away-side observed in heavy-ion collisions at RHIC [30–35].

In this paper we present measurements of the system size and collision energy dependence of near-side di-hadron correlations using data from Cu+Cu and Au+Au collisions at  $\sqrt{s_{NN}} = 62.4$  GeV and  $\sqrt{s_{NN}} = 200$  GeV measured by the STAR experiment at RHIC. In particular, we investigate the centrality dependence of the jet-like correlations and the ridge and the transverse momentum dependence of the jet-like correlations. The properties of jet-like correlations in heavy-ion collisions are compared to those from d+Au collisions and PYTHIA simulations to look for possible medium modifications and broadening of the near-side jet-like correlation e.g. due to gluon bremsstrahlung [17]. The new results on the system size and energy dependence of the ridge yield presented in this article extend our knowledge of this phenomenon and, in combination with other measurements at RHIC, provide important quantitative input and constraints to model calculations.

## II. EXPERIMENTAL SETUP AND DATA SAMPLE

The results presented below are based on data measured by the STAR experiment from d+Au collisions at  $\sqrt{s_{NN}} = 200$  GeV in 2003, Au+Au collisions at  $\sqrt{s_{NN}} = 62.4$  GeV and 200 GeV in 2004, and Cu+Cu collisions at  $\sqrt{s_{NN}} = 62.4$  GeV and 200 GeV in 2005. The d+Au events were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located at 18 m from the nominal interaction point in the Au beam direction, accepting  $95 \pm 3\%$  of the Au+Au hadronic cross section [36]. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) at forward rapidity ( $3.3 < |\eta| < 5.0$ ) and a coincidence between the ZDCs. The MB trigger for Au+Au collisions at  $\sqrt{s_{NN}} = 62.4$  GeV and 200 GeV was obtained using a ZDC coincidence, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions.

For Au+Au collisions at  $\sqrt{s_{NN}} = 200$  GeV, an additional online trigger for central collisions was used. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB. The central trigger sampled the most central 12% of the total hadronic cross section.

In order to achieve a more uniform detector acceptance, only those events with the primary vertex position along the longitudinal beam direction ( $z$ ) within 30 cm of the center of the STAR detector were used for the analysis. For the d+Au collisions this was expanded to  $|z| < 50$  cm.

The STAR Time Projection Chamber (TPC) [37] was used for tracking of charged particles. The collision centrality was determined from the uncorrected number of charged tracks at mid-rapidity ( $|\eta| < 0.5$ ) in the TPC. The charged tracks used for the centrality determination had a three dimensional distance of closest approach (DCA) to the primary vertex of less than 3 cm and at least 10 fit points from the TPC. Each data sample was then divided into several centrality bins, and the fraction of the geometric cross section, the number of participating nucleons ( $N_{part}$ ) and the number of binary collisions

( $N_{coll}$ ) were calculated using Glauber Monte-Carlo model calculations [38].

## III. DATA ANALYSIS

### A. Correlation technique

For the analysis presented in this paper we used charged particle tracks reconstructed in the TPC. Tracks were required to have at least 15 fit points in the TPC, a DCA to the primary vertex of less than 1 cm, and a pseudorapidity  $|\eta| < 1.0$ . As in previous di-hadron correlation studies in the STAR experiment [12, 39, 40], a high- $p_T$  trigger particle was selected and the raw distribution of associated tracks relative to that trigger in pseudorapidity ( $\Delta\eta$ ) and azimuth ( $\Delta\phi$ ) is studied. This distribution,  $d^2N_{raw}/d\Delta\phi d\Delta\eta$ , is normalized by the number of trigger particles,  $N_{trigger}$ , and corrected for the efficiency and acceptance of associated tracks  $\epsilon$ :

$$\frac{d^2N}{d\Delta\phi d\Delta\eta}(\Delta\phi, \Delta\eta) = \frac{1}{N_{trigger}} \frac{d^2N_{raw}}{d\Delta\phi d\Delta\eta} \frac{1}{\epsilon_{assoc}(\phi, \eta)} \frac{1}{\epsilon_{pair}(\Delta\phi, \Delta\eta)}. \quad (1)$$

The efficiency correction  $\epsilon_{assoc}(\phi, \eta)$  is a correction for the TPC single charged track reconstruction efficiency and  $\epsilon_{pair}(\Delta\phi, \Delta\eta)$  is a correction for track merging and finite TPC track-pair acceptance in  $\Delta\phi$  and  $\Delta\eta$  as described in detail below. The data presented in this paper are averaged between positive and negative  $\Delta\phi$  and  $\Delta\eta$  regions and are reflected about  $\Delta\phi = 0$  and  $\Delta\eta = 0$  in the plots.

### B. Single charged track efficiency correction

The single charged track reconstruction efficiency in the TPC is determined by simulating the detector response to a charged particle and embedding these signals into a real event. This hybrid event is analyzed using the same software as for the real events. The efficiency for detecting a single track as a function of  $p_T$ ,  $\eta$ , and centrality is determined from the number of simulated particles which were successfully reconstructed. The single track efficiency is approximately constant for  $p_T > 2$  GeV/c and ranges from around 75% for central Au+Au events to around 85% for peripheral Cu+Cu events as shown in Figure 1. The efficiency for reconstructing a track in d+Au is 89%. The systematic uncertainty on the efficiency correction, 5%, is strongly correlated across centralities and  $p_T$  bins for each data set but not between data sets. In the correlations, each track pair is corrected for the efficiency for reconstructing the associated particle. Since the correlations are normalized by the number of trigger particles, no correction for the efficiency of the trigger particle is necessary.

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

| System | Centrality | $\sqrt{s_{NN}}$ [GeV] | No. of events [ $10^6$ ] |
|--------|------------|-----------------------|--------------------------|
| Cu+Cu  | 0-60%      | 62.4                  | 24                       |
| Au+Au  | 0-80%      | 62.4                  | 8                        |
| d+Au   | 0-95%      | 200                   | 3                        |
| Cu+Cu  | 0-60%      | 200                   | 38                       |
| Au+Au  | 0-80%      | 200                   | 28                       |
| Au+Au  | 0-12%      | 200                   | 17                       |



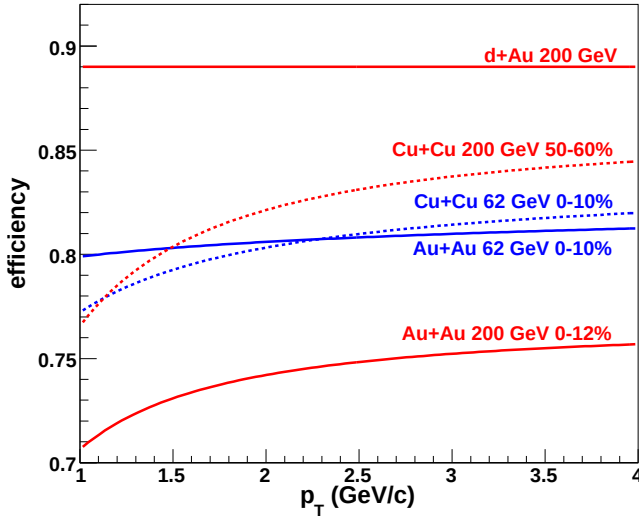


FIG. 1: Parameterizations of the transverse momentum dependence of the reconstruction efficiency of charged particles in the TPC in various collision systems, energies and centrality bins for the track selection cuts used in this analysis. Note the zero suppression of the axes. Color online.

### C. Corrections for track merging and track crossing effects in the TPC

The reconstruction of charged tracks from TPC hits is performed iteratively, with hits removed from the event once they are assigned to a track. If two tracks have small angular separation in both pseudorapidity and azimuth, they are more difficult to reconstruct because distinct hits from each particle may not be resolved by the TPC. If two particles are close in momenta or have sufficiently high  $p_T$  that their tracks are nearly straight, they may not be distinguished. This effect, called track merging, reduces the number of pairs observed at small opening angles and results in an artificial dip in the raw correlations centered at  $(\Delta\phi, \Delta\eta) = (0, 0)$ .

Figure 2 shows an example of a  $(\Delta\phi, \Delta\eta)$  two-particle di-hadron correlation function in central Au+Au collisions at  $\sqrt{s_{NN}} = 200$  GeV, along with the corresponding  $\Delta\phi$  and  $\Delta\eta$  projections for  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5$  GeV/c  $< p_T^{\text{associated}} < p_T^{\text{trigger}}$ . For small angular separations, a clear dip in the raw correlations is visible and must be corrected for in order to extract the yield of associated particles on the near side.

Another similar effect is evident in the active TPC volume from high- $p_T$  tracks which cross. While few hits are lost in this case, one track may lose hits near the crossing point and therefore be split into two shorter tracks. Shorter tracks are less likely to meet the track selection criteria. Track merging and track crossing cause four dips near  $(\Delta\phi, \Delta\eta) = (0, 0)$  but slightly displaced in  $\Delta\phi$ . The location and width of these dips in  $\Delta\phi$  is dependent on the relative helicities,  $h$ , and the  $p_T$  intervals of the trigger and associated particles. The helicity  $h$  is given

by

$$h = \frac{-qB}{|qB|} \quad (2)$$

where  $q$  is the charge of the particle and  $B$  is the magnetic field. The dips for tracks of the same helicity are dominantly due to track merging and the dips for tracks of opposite helicities are dominantly due to track crossing. Figure 3 displays the correlation function from Figure 2 in four different helicity combinations of trigger and associated particles showing the finer substructure of the dip on the near side. When the helicities of the trigger and associated particles are the same, the percentage of overlapping hits is greater. Because higher  $p_T$  tracks have a smaller curvature, it is more likely for two high  $p_T$  tracks close in azimuth and pseudorapidity to be merged than lower  $p_T$  particles. However, track pairs are lost whether the pair is part of the combinatorial background or part of the signal. This effect means that the magnitude of the dip is greater in central collisions where the background is greater and decreases with increasing  $p_T$  because of the decreasing background.

One way to correct for the track merging effect is to remove pairs from mixed events that would have merged in real data. The environment in mixed events must be similar to real data in order for pair rejection to be accurately reproduced. The reference multiplicities of both events were required to be within 10 of each other to assure similar track density. In addition, mixed events were required to have vertices within 2 cm of each other along the beam axis in order to ensure similar geometric acceptance to avoid a different dip shape in  $\Delta\eta$ . In order to calculate accurately the percentage of merged hits, the origin of the associated track was shifted to the vertex of the event which the trigger particle originated from.

Previous analyses of low momentum tracks have shown that eliminating pairs from both data and mixed events with a fraction of merged hits greater than 10% was sufficient to correct for merging [41]. By discarding pairs with more than 10% shared hits we insure that the percentage of merged track pairs is the same in the data and the mixed events.

The correlation function for a given helicity combination of trigger and associated particles was corrected by mixed events. After this correction, a small residual dip remains, mostly due to track crossing. While the mixed events correct for the dip due to true track merging well, they do not correct for track crossing as well. The remaining dip is then corrected for using the symmetry of the correlations. Since the data are symmetric about  $\Delta\phi = 0$ , the data on the same side as the dip are discarded and replaced by the data on the side without the dip. Then the data are reflected about  $\Delta\eta = 0$  and added to the unreflected data to minimize statistical fluctuations. This method is only applied to  $|\Delta\phi| < 1.05$  and  $|\Delta\eta| < 0.67$ , the region shown in Figure 2, because it is computationally intensive and track merging and track

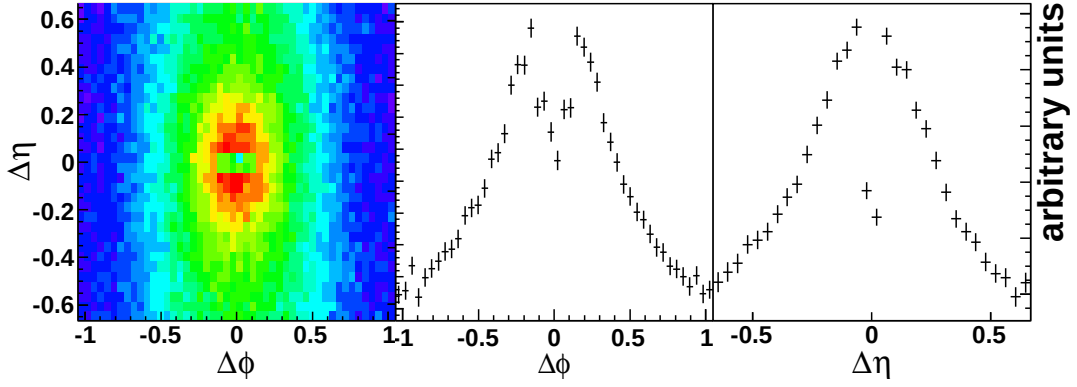


FIG. 2: Two-particle di-hadron correlation function in  $(\Delta\phi, \Delta\eta)$  for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$  and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  in 0-12% central Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ , not corrected for track merging (left). The projection of the correlation function in  $\Delta\phi$  for  $|\Delta\eta| < 0.042$  is shown in the middle and the projection in  $\Delta\eta$  for  $|\Delta\phi| < 0.17$  on the right. Color online.

crossing only affect small angular separations. For large  $\Delta\phi$  and  $\Delta\eta$ , the method described in the next section is applied. The track merging correction is done for each di-hadron correlation function separately with the appropriate cuts on  $p_T^{\text{trigger}}$ ,  $p_T^{\text{associated}}$  and collision centrality. An example of the di-hadron correlation function before and after the track merging correction is shown in Fig-

ure 4.

#### D. Pair acceptance correction

With the restriction that each track falls within  $|\eta| < 1.0$ , there is a limited acceptance for track pairs. For  $\Delta\eta \approx 0$ , the geometric acceptance of the TPC for track pairs is  $\approx 100\%$ , however, near  $\Delta\eta \approx 2$  the acceptance is close to 0%. In azimuth the acceptance is limited by the 12 TPC sector boundaries, leading to dips in the acceptance of track pairs in azimuth. To correct for the geometric acceptance, the distribution of tracks as a function of  $\eta$  and  $\phi$  was recorded for both trigger and associated particles. A random  $\eta$  and  $\phi$  was chosen from each of these distributions to reconstruct a random  $\Delta\eta$  and  $\Delta\phi$ . This was done for at least four times as many track pairs as in the data and was used to calculate the geometrical acceptance correction for pairs.

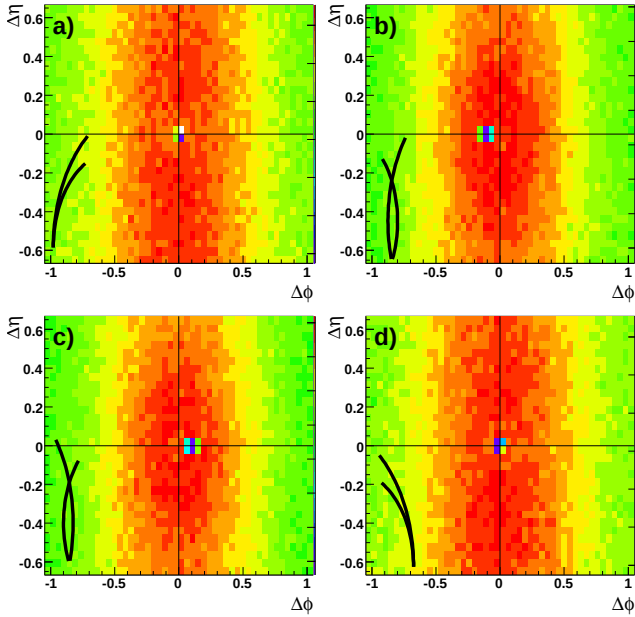


FIG. 3: The dip region in  $(\Delta\phi, \Delta\eta)$  uncorrected di-hadron correlations in 0-12% central Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$  for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$  and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  in the four helicity combinations of trigger and associated particles: (a)  $(h_{\text{trig}}, h_{\text{assoc}}) = (1,1)$ , (b)  $(h_{\text{trig}}, h_{\text{assoc}}) = (1,-1)$ , (c)  $(h_{\text{trig}}, h_{\text{assoc}}) = (-1,1)$ , and (d)  $(h_{\text{trig}}, h_{\text{assoc}}) = (-1,-1)$ . Cartoons indicate which dips are from track merging and which are from track crossing. Color online.

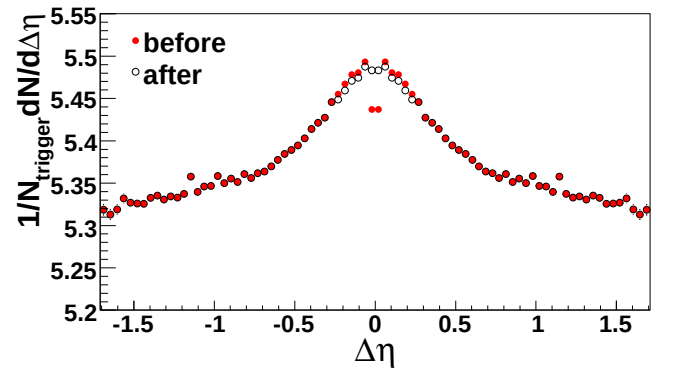


FIG. 4: The raw correlation in  $\Delta\eta$  for di-hadron correlations for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$  and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  for 0-12% central Au+Au collisions for  $|\Delta\phi| < 0.78$  before and after the track merging correction is applied. The data have been reflected about  $\Delta\eta = 0$ . Color online.

## E. Subtraction of anisotropic elliptic flow background

Correlations of particles with the reaction plane due to anisotropic flow ( $v_2$ ) in heavy-ion collisions are indirectly reflected in di-hadron correlations and have to be subtracted for studies of the ridge. This background in  $\Delta\phi$  over the interval  $[-a, a]$  is approximated by

$$B_{\Delta\phi}[-a, a] \equiv b_{\Delta\phi} \int_{-a}^a d\Delta\phi \left( 1 + 2\langle v_2^{\text{trig}} \rangle \langle v_2^{\text{assoc}} \rangle \cos 2\Delta\phi \right) \quad (3)$$

where  $a$  is chosen to be 0.78 so that the entire signal is contained. The level  $b_{\Delta\phi}$  of the background is determined using the Zero Yield At Minimum (ZYAM) method [8]. The level of the background is taken as the value of the minimum bin. The systematic errors due to the choice of the minimum bin rather than either of the two neighboring bins are negligible compared to the systematic errors due to the magnitude of  $v_2$ , discussed below.

The ZYAM method is commonly used for di-hadron correlations at RHIC, for example [8, 11, 42], and is justified if the near- and away-side peaks are separated by a 'signal-free' region. At lower transverse momenta ( $p_T < 2$  GeV/c) and in central Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV the broadening of the away-side correlation peak may cause overlap of the near- and away-side peaks and consequently makes the ZYAM normalization procedure biased. Alternatively, a decomposition of the correlation function using a fit function containing the anisotropic flow modulation of the combinatorial background and components describing the shape of the correlation peaks could be used as well [43, 44]. In this paper we use the ZYAM prescription to remain consistent with our earlier measurements of the near-side ridge [12]. ZYAM used with conservative bounds on  $v_2$  will, if anything, underestimate the ridge yield.

For all collision systems and energies studied, the uncertainty bounds on  $v_2$  were determined by comparing different methods for the  $v_2$  measurement. We assume that the error on the  $v_2$  of the trigger and associated particles is 100% correlated. Event plane measurements of flow and two-particle measurements such as the two-particle cumulant method are sensitive to non-flow from sources such as jets. These methods may overestimate  $v_2$ . Methods such as the four-particle cumulant method are less sensitive to contributions from jets, however, these methods may over-subtract contributions from event-by-event fluctuations in  $v_2$ . Therefore these methods underestimate the  $v_2$  that should be used for the background subtraction in di-hadron correlations [45]. For each system at least one measurement which may overestimate  $v_2$  and at least one measurement which may underestimate  $v_2$  is included.  $v_2$  and systematic errors on  $v_2$  for Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV are from comparisons of the event plane method using the forward

TPCs for the event plane determination and the four particle cumulant method [46]. The  $v_2$  and systematic errors on  $v_2$  for Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV are as described in [12]. The upper bound on  $v_2$  is from the reaction plane method using the forward TPCs for the determination of the reaction plane, the lower bound comes from the four particle cumulant method, and the average of the two is the nominal value.  $v_2$  for Cu+Cu collisions at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV and 200 GeV is from [47]. The nominal value is given by  $v_2$  from the reaction plane method using the forward TPCs for the determination of the reaction plane and the upper bound is from the statistical error for both  $\sqrt{s_{\text{NN}}} = 62.4$  GeV and 200 GeV. For Cu+Cu collisions measurements using the four particle cumulant method were not possible due to limited statistics. Instead, for Cu+Cu collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV the lower bound is determined by calculating the  $v_2$  from p+p collisions, scaling it by  $N_{\text{coll}}$ , and subtracting it from  $v_2$  determined using the reaction plane method to estimate the maximum contribution from nonflow in Cu+Cu collisions. For Cu+Cu collisions at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV the systematic error is assumed to be the same in Cu+Cu collisions at both energies.

With both methods for subtracting the ridge contribution to the jet-like yield described below, the systematic errors due to  $v_2$  cancel out in the jet-like yield, assuming that  $v_2$  is independent of  $\Delta\eta$  in the TPC acceptance. This assumption is based on the measurements of  $v_2$  as a function of  $\eta$  [48, 49].

## F. Yield extraction

To quantify the strength of the near-side correlation it is assumed that it can be decomposed into a jet-like component, narrow in both azimuth and pseudorapidity, and a ridge component which is independent of  $\Delta\eta$ . This approach is consistent with the method in [12]. For the kinematic cuts applied to  $p_T^{\text{trigger}}$  and  $p_T^{\text{associated}}$ , the jet-like correlation is contained within the cuts used in this analysis,  $|\Delta\eta| < 0.78$  and  $|\Delta\phi| < 0.78$ .

To study the jet-like correlation and the ridge quantitatively we adopt the notation from [12], and introduce the projection of the di-hadron correlation function from Eq. (1) onto the  $\Delta\eta$  axis:

$$\left. \frac{dN}{d\Delta\eta} \right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}; \quad (4)$$

and similarly on the  $\Delta\phi$  axis:

$$\left. \frac{dN}{d\Delta\phi} \right|_{a,b} \equiv \int_{|\Delta\eta| \in [a,b]} d\Delta\eta \frac{d^2N}{d\Delta\phi d\Delta\eta}. \quad (5)$$

To determine the jet-like yield of associated charged particles two methods are used. The first method is based

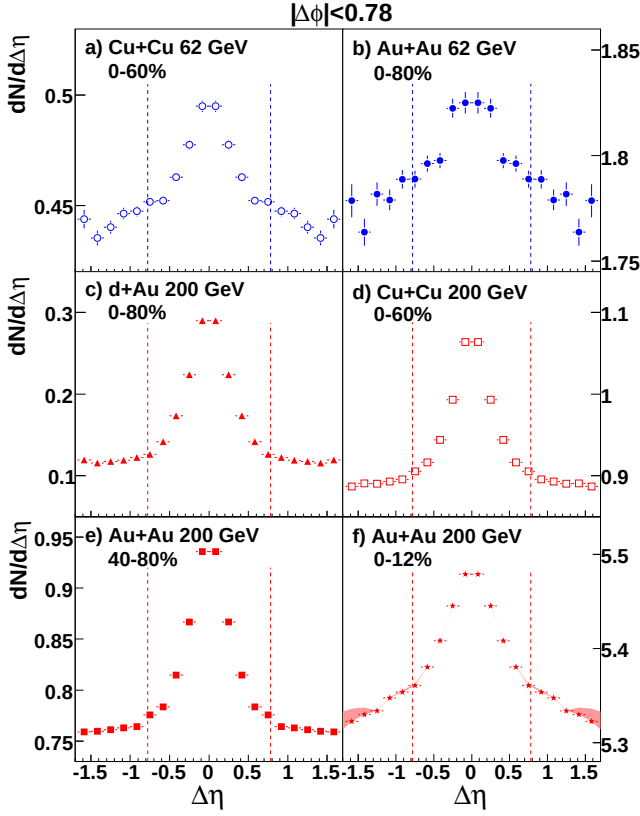


FIG. 5: Sample correlations in  $\Delta\eta$  ( $|\Delta\phi| < 0.78$ ) for  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5$  GeV/c  $< p_T^{\text{associated}} < p_T^{\text{trigger}}$  for (a) 0-60% Cu+Cu at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV (b) 0-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV, (c) minimum bias Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV, (d) 0-60% Cu+Cu at  $\sqrt{s_{\text{NN}}} = 200$  GeV, (e) 40-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV, and (f) 0-12% central Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV. Lines show the  $\Delta\eta$  range where the jet-like yield is determined. The data are averaged between positive and negative  $\Delta\eta$  and reflected in the plot. Shaded bands in (f) show the systematic errors discussed in III F. Color online.

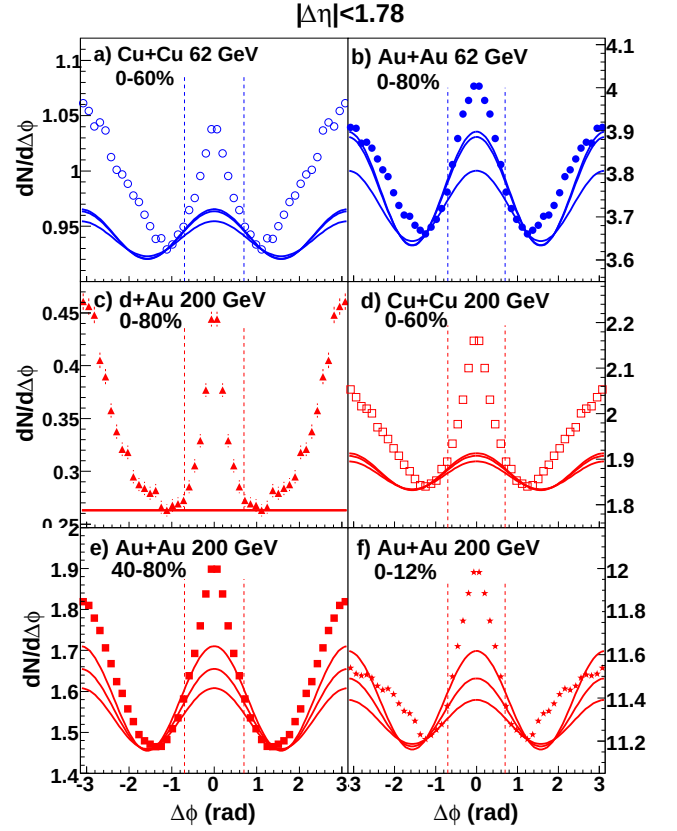


FIG. 6: Sample correlations in  $\Delta\phi$  ( $|\Delta\eta| < 1.78$ ) for  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5$  GeV/c  $< p_T^{\text{associated}} < p_T^{\text{trigger}}$  for (a) 0-60% Cu+Cu at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV (b) 0-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV, (c) minimum bias Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV, (d) 0-60% Cu+Cu at  $\sqrt{s_{\text{NN}}} = 200$  GeV, (e) 40-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV, and (f) 0-12% central Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV. Solid lines show the estimated background using the ZYAM method with the range of  $v_2$  used for the determination of the systematic errors. The data are averaged between positive and negative  $\Delta\phi$  and reflected in the plot. Vertical dashed lines show the  $\Delta\phi$  range where the jet-like correlation is determined. Color online.

on  $\Delta\phi$  projections. Under the assumption that the jet-like yield is confined within  $|\Delta\eta| < 0.78$ , subtracting the  $\Delta\phi$  projections:

$$\frac{dN_J}{d\Delta\phi}(\Delta\phi) = \frac{dN}{d\Delta\phi} \Big|_{0,0.78} - \frac{0.78}{1.0} \frac{dN}{d\Delta\phi} \Big|_{0.78,1.78} \quad (6)$$

removes both the elliptic flow and ridge contributions. Since the second  $\Delta\phi$  projection is calculated in a larger  $\Delta\eta$  window, it has to be scaled by a factor  $0.78/1.0$ , the ratio of the  $\Delta\eta$  width in the region containing the jet-like correlation, the ridge, and the background to the width of the region containing only the ridge and the background. This subtracts both the ridge and  $v_2$  simultaneously since both are independent of  $\Delta\eta$ .

The jet-like yield  $Y_J^{\Delta\phi}$  is then obtained by integrating

Eq.(6)

$$Y_J^{\Delta\phi} = \int_{-0.78}^{0.78} d\Delta\phi \frac{dN_J}{d\Delta\phi}(\Delta\phi), \quad (7)$$

The second method for jet-like yield determination is based on the  $\Delta\eta$  projection at the near-side:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \frac{dN}{d\Delta\eta} \Big|_{-0.78,0.78} - b_{\Delta\eta} \quad (8)$$

as  $v_2$  is independent of pseudo-rapidity within the STAR acceptance and therefore only leads to a constant offset included in  $b_{\Delta\eta}$ . The background level  $b_{\Delta\eta}$  is determined by fitting a constant background  $b_{\Delta\eta}$  plus a Gaussian to  $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$ . The yield determined from fit is discarded to avoid any assumptions about the shape of the peak and



instead we integrate Eq. (8) over  $\Delta\eta$  using bin counting to determine the jet-like yield  $Y_J^{\Delta\eta}$ :

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \frac{dN_J}{d\Delta\eta}(\Delta\eta), \quad (9)$$

The ridge yield  $Y_{\text{ridge}}$  is determined by first evaluating Eq. (5) over the entire  $\Delta\eta$  region to get  $\frac{dN}{d\Delta\phi}$  then subtracting the modulated elliptic flow background  $B_{\Delta\phi}$  and the jet-like contribution  $Y_J^{\Delta\eta}$ :

$$Y_{\text{ridge}} = \int_{-0.78}^{0.78} d\Delta\phi \left. \frac{dN}{d\Delta\phi} \right|_{0,1.78} - B_{\Delta\phi}[-0.78, 0.78] - Y_J^{\Delta\eta}. \quad (10)$$

We determined the systematic error on the  $Y_J$  due to uncertainty in the acceptance correction by comparing the mixed event method described in III D to the standard event mixing method and to a sample with a restricted z vertex position. The largest difference was seen in the central Au+Au data at  $\sqrt{s_{\text{NN}}} = 200$  GeV. To be conservative this difference is used as the systematic error for all the data sets. The resulting systematic errors are asymmetric and the reported values for  $Y_J$  are upper bounds. For  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5$  GeV/c  $< p_T^{\text{associated}} < p_T^{\text{trigger}}$  and for higher momentum trigger and associated particles this systematic error is 9%. For  $1.5$  GeV/c  $< p_T^{\text{associated}} < p_T^{\text{trigger}}$  the systematic error for  $2.0 < p_T^{\text{trigger}} < 2.5$  GeV/c is 27% and for  $2.5 < p_T^{\text{trigger}} < 3.0$  GeV/c is 18%. For  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.0 < p_T^{\text{associated}} < p_T^{\text{trigger}}$  GeV/c the systematic error is 16%. This error is also present for the ridge, since the ridge is determined by subtracting  $Y_J$ . The systematic error on the widths determined in the same way does not exceed 6% for  $p_T^{\text{trigger}} > 2.5$  GeV/c and 10% for  $p_T^{\text{trigger}} < 2.5$  GeV/c. Additionally, the systematic error on  $Y_J$  due to the track merging correction does not exceed 1%, the maximum size of the correction in the kinematic region studied in this paper. This correction does not impact  $Y_{\text{ridge}}$ .

### G. 2D Fits

In addition to the standard ZYAM procedure, we also analyzed the distribution of particles in Eq. (1) using two-dimensional fits of the form:

$$\frac{d^2N}{d\Delta\phi d\Delta\eta} = A \left( 1 + \sum_{n=1}^4 2\tilde{v}_n^2 \cos(n\Delta\phi) \right) + \frac{Y_J}{2\pi\sigma_{\Delta\phi,J}\sigma_{\Delta\eta,J}} e^{-\frac{\Delta\eta}{2\sigma_{\Delta\eta,J}^2}} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi,J}^2}}. \quad (11)$$

with first four coefficients  $\tilde{v}_1^2$ ,  $\tilde{v}_2^2$ ,  $\tilde{v}_3^2$ , and  $\tilde{v}_4^2$  of a Fourier expansion and a term accounting for the jet-like correlation on the near-side. This approach is motivated by

the class of models for ridge production through a triangular initial condition. If non-flow contributions are negligible,  $\tilde{v}_2^2$  corresponds to the average of the product of the trigger particle  $v_2$  and the associated particle  $v_2$  and  $\tilde{v}_3^2$  corresponds to the average of the product of the trigger particle  $v_3$  and the associated particle  $v_3$ . We use Eq. (11) to fit the data and extract  $(\tilde{v}_3/\tilde{v}_2)^2$  for all collision energies and systems. We allow  $\tilde{v}_3^2$  to be negative. A narrow roughly Gaussian away-side peak at  $\Delta\phi \approx \pi$ , which could arise from correlations from the production of an away-side jet, would lead to a negative  $\tilde{v}_3^2$  and would indicate that flow is not the dominant production mechanism for these correlations. Furthermore,  $\tilde{v}_2$  is not constrained to the experimental values measured for  $v_2$  through other means.

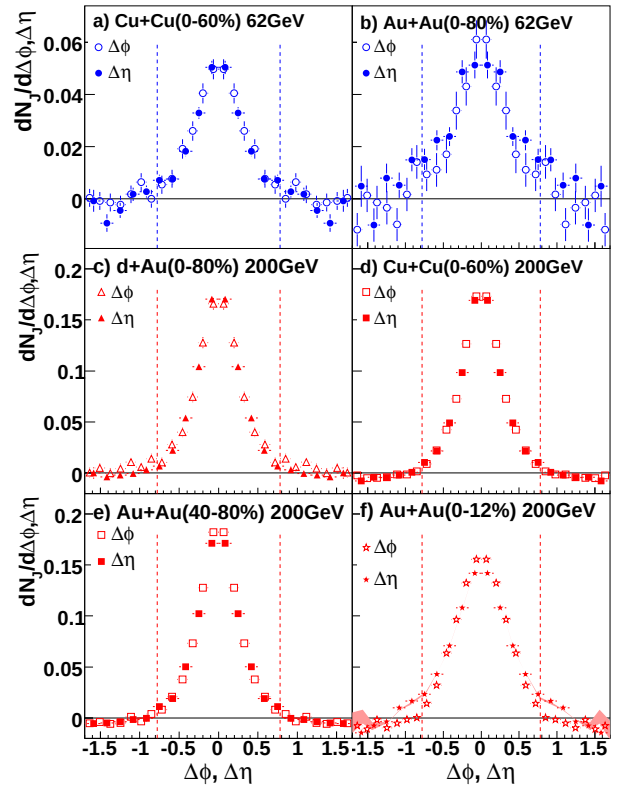


FIG. 7: Background subtracted sample correlations for  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5$  GeV/c  $< p_T^{\text{associated}} < p_T^{\text{trigger}}$  on the near-side for (a) 0-60% Cu+Cu at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV, (b) 0-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 62.4$  GeV, (c) minimum bias d+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV, (d) 0-60% Cu+Cu at  $\sqrt{s_{\text{NN}}} = 200$  GeV, (e) 40-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV and (f) 0-12% central Au+Au at  $\sqrt{s_{\text{NN}}} = 200$  GeV. The dependence of the jet-like correlation is shown as a function of  $\Delta\phi$  ( $|\Delta\eta| < 1.78$ ) in open symbols and of  $\Delta\eta$  ( $|\Delta\phi| < 0.78$ ) in closed symbols. Lines show the  $\Delta\phi$  and  $\Delta\eta$  ranges where the jet-like yield is determined. The data are averaged between positive and negative  $\Delta\eta$  ( $\Delta\phi$ ) and reflected in the plot. Shaded bands in (f) show the systematic errors on  $\frac{dN_J}{d\Delta\eta}$  discussed in III F. Color online.

## IV. RESULTS

### A. Sample Correlations

Figure 5 shows fully corrected  $\Delta\eta$  projections of sample correlations on the near side ( $|\Delta\phi| < 0.78$ ) before background subtraction for d+Au, Cu+Cu and Au+Au collisions at energy  $\sqrt{s_{NN}} = 62.4$  GeV and  $\sqrt{s_{NN}} = 200$  GeV. The trigger particles were selected with transverse momentum  $3 < p_T^{\text{trigger}} < 6$  GeV/c and the associated particles with  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$ . The data show a clear jet-like peak sitting on top of the background and the ridge. The level of the background is increasing with energy and system size as expected, as more bulk particles are produced in the collision.

Examples of the complementary projections in  $\Delta\phi$  before background subtraction for ( $|\Delta\eta| < 1.78$ ) are shown in Figure 6. The elliptic flow modulated background is shown as solid curves. The middle curve corresponds to background calculated with the nominal value of  $v_2$ . The upper (lower) curve corresponds to the background if the upper (lower) bound on  $v_2$  is used instead. Since we have conservatively assumed that the error on the  $v_2$  of the trigger and associated particles is 100% correlated, the background occasionally goes above the signal in Figure 6f on the away-side. However, since we are focusing the near-side we prefer this conservative estimate. Note that the uncertainty in the size of the elliptic flow modulated background affects only the ridge yield but not the jet-like yield, since the elliptic flow contribution to the jet-like yield in  $\Delta\phi$  cancels out in Eq. (6) and in  $\Delta\eta$  is included in  $b_{\Delta\eta}$  in Eq. (8).

Sample background subtracted correlation functions  $\frac{dN_J}{d\Delta\eta}$  from Eq.(8) and  $\frac{dN_J}{d\Delta\phi}$  from Eq.(6) on the near side for  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  from Figure 5 and Figure 6 are shown in Figure 7. For the given kinematic selection, the extracted jet-like correlation peaks in both  $\Delta\eta$  and  $\Delta\phi$  projections look very similar in all studied systems and collision energies. The jet-like yields discussed through the rest of the paper are obtained from the  $\Delta\eta$  projection method; the  $\Delta\phi$  method is only used for determining the width of the jet-like correlation in  $\Delta\phi$ . Below, the dependence of the near side jet-like yield and Gaussian width of the jet-like correlation peak on collision centrality and the transverse momentum of the trigger and associated particles are studied in detail.

### B. The near-side jet-like component

The centrality dependence of the jet-like yield for  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  is plotted in Figure 8. The jet-like yield at  $\sqrt{s_{NN}} = 62.4$  GeV is lower than that at  $\sqrt{s_{NN}} = 200$  GeV by about a factor of three, which can be understood as the result of steeply falling jet spectrum folded with the fragmentation

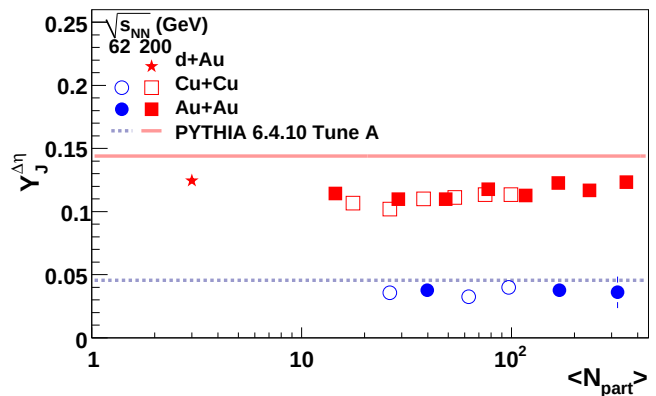


FIG. 8: Dependence of jet-like yields on  $N_{\text{part}}$  for  $3 < p_T^{\text{trigger}} < 6$  GeV/c and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  for Cu+Cu, and Au+Au at  $\sqrt{s_{NN}} = 62.4$  GeV and d+Au, Cu+Cu and Au+Au at  $\sqrt{s_{NN}} = 200$  GeV. Comparisons to PYTHIA,  $N_{\text{part}} = 2$ , are shown as shaded bands. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown. The background level and  $v_2$  values used for the extraction of these yields are given in Table III. Color online.

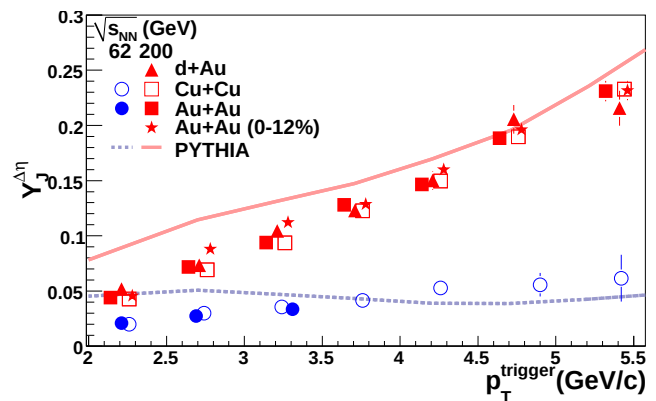


FIG. 9: Dependence of jet-like yield on  $p_T^{\text{trigger}}$  for minimum bias d+Au, 0-60% Cu+Cu at  $\sqrt{s_{NN}} = 62.4$  GeV and  $\sqrt{s_{NN}} = 200$  GeV, 0-80% Au+Au at  $\sqrt{s_{NN}} = 62.4$  GeV, and 0-12% and 40-80% Au+Au at  $\sqrt{s_{NN}} = 200$  GeV. Comparisons to PYTHIA are shown as shaded bands. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown. Color online.

function. For a given number of participating nucleons,  $N_{\text{part}}$ , and collision energy,  $\sqrt{s_{NN}}$ , there is no significant difference between the d+Au, Cu+Cu and Au+Au collisions observed, as expected if the jet-like correlation were dominantly produced by vacuum fragmentation. The measured yields are also compared to PYTHIA simulations shown as a band in Figure 8. For these studies PYTHIA version 6.4.10 [50] CDF tune A [51], which matches the data from the Tevatron at  $\sqrt{s} = 1.8$  TeV and also describes the pion and proton inclusive spectra well at RHIC energies [52–54], is used. The PYTHIA prediction is somewhat above the data, even for d+Au collisions. However, considering the fact that the data are

from heavy-ion collisions and the PYTHIA simulations are for p+p collisions, good agreement is unanticipated.

The dependence of the jet-like yield on  $p_T^{\text{trigger}}$  for  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  is plotted in Figure 9 for all studied collision systems and energies. Data from Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$  are shown separately for peripheral (40-80%) and central (0-12%) Au+Au collisions. The jet-like yield increases constantly with  $p_T^{\text{trigger}}$  for both Cu+Cu and Au+Au and for  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  and  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . The effect of a steeper jet spectrum at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  relative to 200 GeV discussed above is now reflected in the difference between the jet-like yields at the two energies. This difference in the jet-like yield between the two studied collision energies increases with  $p_T^{\text{trigger}}$  from about a factor of two at  $p_T^{\text{trigger}} = 2.5 \text{ GeV}/c$  to a factor of four at  $p_T^{\text{trigger}} = 5.5 \text{ GeV}/c$ . No significant differences between d+Au, Cu+Cu and Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$  are observed. This finding is consistent with the jet-like correlation arising from fragmentation. Comparisons to PYTHIA simulations are shown as lines in Figure 9. It is surprising how well PYTHIA is able to describe the  $p_T^{\text{trigger}}$  dependence of the jet-like yield in A+A collisions. In general, the agreement is better at larger  $p_T^{\text{trigger}}$  ( $p_T^{\text{trigger}} > 4 \text{ GeV}/c$ ), while at lower  $p_T^{\text{trigger}}$  values PYTHIA predicts a larger jet-like yield than observed in the data.

The spectra of particles associated with the jet-like correlation and their comparison to PYTHIA simulations for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$  are shown in Figure 10. For the same  $p_T^{\text{trigger}}$  selection, the mean transverse momentum fraction  $z_T$  carried by the leading hadron is larger at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  than at 200 GeV due to the steeper jet spectrum. This is reflected in softer  $p_T^{\text{associated}}$  spectra at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$ . The inverse slope parameters from an exponential fit to these data are shown in Table II. There is no difference seen between Cu+Cu and peripheral Au+Au in either the data points or the extracted inverse slope parameter. The inverse slope parameter of the central Au+Au data at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$  is somewhat lower than the other data at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ , largely because of the larger yield at the lowest  $p_T^{\text{associated}}$ .

TABLE II: Inverse slope parameters in MeV/c of  $p_T^{\text{associated}}$  spectra from fits of data in Figure 10. The inverse slope parameter from a fit to  $\pi^-$  inclusive spectra in Au+Au collisions [55, 56] above 1.0 GeV/c is  $280.9 \pm 0.4 \text{ MeV}/c$  for 0-10%  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  and  $330.9 \pm 0.3 \text{ MeV}/c$  for 0-12%  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . Statistical errors only.

|        | $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$ | $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$    |
|--------|-------------------------------------------|---------------------------------------------|
| Au+Au  | $332 \pm 13$                              | $457 \pm 4$ (40-80%)<br>$399 \pm 4$ (0-12%) |
| Cu+Cu  | $370 \pm 9$                               | $443 \pm 3$                                 |
| d+Au   |                                           | $438 \pm 9$                                 |
| PYTHIA | $417 \pm 9$                               | $491 \pm 3$                                 |

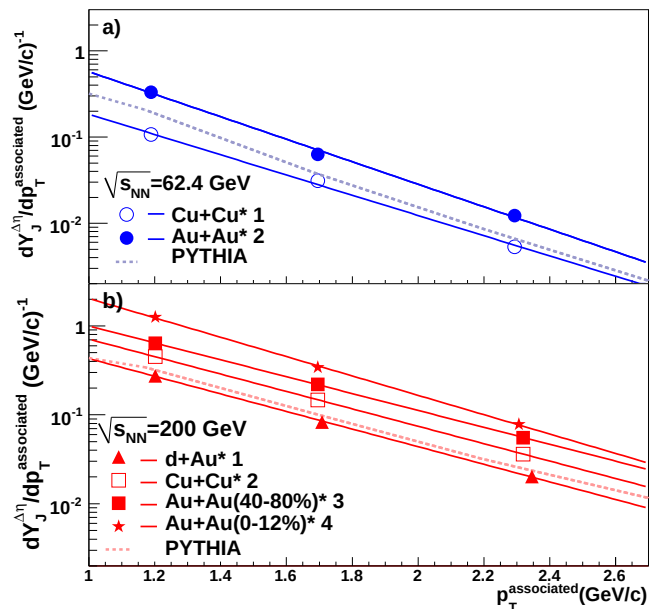


FIG. 10: Dependence of jet-like yield on  $p_T^{\text{associated}}$  for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$  for (a) 0-60% Cu+Cu and 0-80% Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  and (b) minimum bias d+Au, 0-60% Cu+Cu, 0-12% Au+Au and 40-80% Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . Solid lines through the data points are fits to the data. Comparisons to PYTHIA are shown as dashed lines. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown. Color on-line.

This also indicates that there is some modification of the jet-like correlation at low  $p_T$ . While the agreement with PYTHIA is remarkable for a comparison to A+A collisions, the discrepancies between PYTHIA and the data are larger at lower momenta and lower energy. This is expected since PYTHIA is tuned better at higher  $p_T$  and higher energy.

Figure 11 shows the Gaussian widths of the  $\Delta\phi$  and  $\Delta\eta$  projections of the near-side jet-like peak as a function of  $p_T^{\text{trigger}}$ ,  $p_T^{\text{associated}}$ , and  $N_{\text{part}}$  along with PYTHIA simulations. There are no significant differences between collision systems except for central Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . While no dependence on collision system is observed, there is a clear increase in the  $\Delta\eta$  width with increasing  $N_{\text{part}}$  in Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . This indicates that the shape of the jet-like correlation is modified in central Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . PYTHIA predicts a greater width in  $\Delta\eta$  at the lowest  $p_T^{\text{associated}}$  than seen in d+Au or Cu+Cu.

Overall it can be concluded that the agreement between the different collision systems and energies shows remarkably little dependence of the jet-like per trigger yield on the system size. In contrast to the peripheral Au+Au data, the central Au+Au data show indications that the jet-like correlation is modified. The model in [57], a hypothesis for the formation of the ridge through gluon bremsstrahlung, does not produce a ridge

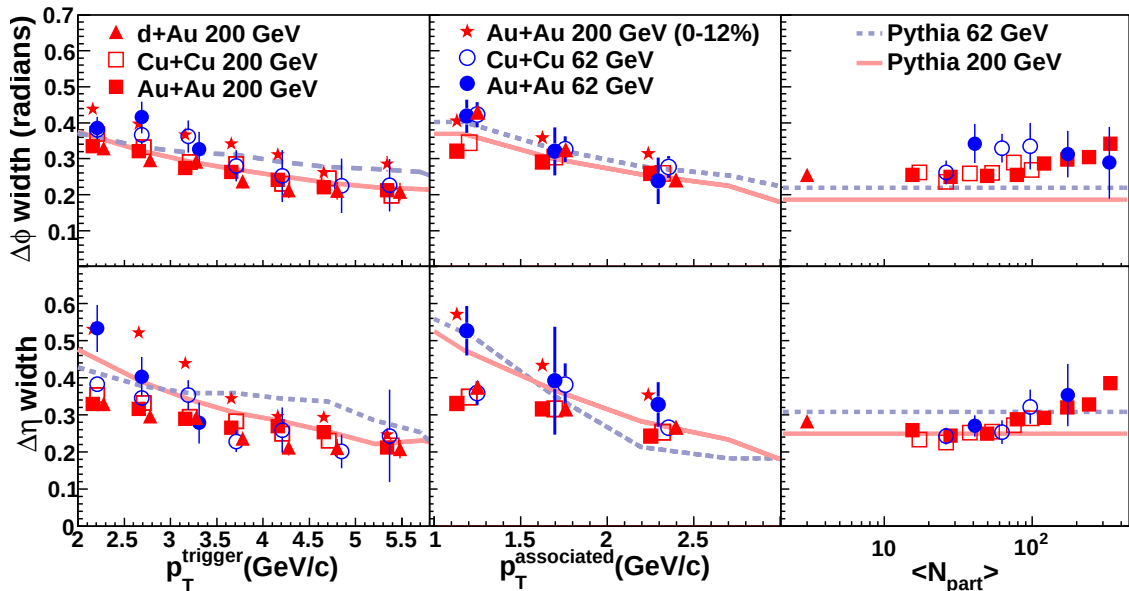


FIG. 11: Dependence of the widths in  $\Delta\phi$  and  $\Delta\eta$  on  $p_T^{\text{trigger}}$  for  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$ ,  $p_T^{\text{associated}}$  for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$ , and  $N_{\text{part}}$  for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$  and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  for minimum bias d+Au, 0-60% Cu+Cu at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  and  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ , 0-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$ , and 0-12% and 40-80% Au+Au at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . Comparisons to PYTHIA are shown as shaded bands. The 5% systematic error due to the uncertainty on the acceptance correction is not shown. Color online.

broad enough to agree with the data, however, it is possible that a similar mechanism could explain the broadening of the jet-like correlation. Similarly models for ridge production by turbulent color fields [26, 58] predict a broadening of the jet-like peak in  $\Delta\eta$  which is not wide enough to describe the ridge but may explain the data in Figure 11.

### C. The near-side ridge

In [12], we reported detailed studies of the ridge in Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$  as a function of  $p_T^{\text{trigger}}$  and  $p_T^{\text{associated}}$ . Here we investigate the ridge centrality, energy and system size dependence. The dependence of the ridge yield on  $N_{\text{part}}$  for Cu+Cu and Au+Au collisions is shown in Figure 12 for both energies studied. Table III shows  $b_{\Delta\phi}$  values and  $v_2$  of trigger and associated particles for all collision systems and energies studied in Figure 12. The centrality bins are characterized by the fraction of geometric cross section  $\sigma/\sigma_{\text{geo}}$ , number of participants  $N_{\text{part}}$  and number of binary collisions  $N_{\text{coll}}$ . Contrary to the jet-like yield, which shows little dependence on centrality, the ridge yield increases steeply with  $N_{\text{part}}$ . Within errors, there is no difference in ridge yield between Cu+Cu and Au+Au collisions at the same  $N_{\text{part}}$  at a given energy, demonstrating the system independence of the ridge yield.

The energy dependence of the ridge yield is potentially a sensitive test of ridge models. Comparing the two collision energies studied, the ridge yield is observed to be

considerably smaller at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  than at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . Similar behavior was also observed for the jet-like yield. Therefore a closer investigation of the centrality dependence of the ratio  $Y_{\text{ridge}}/Y_{\text{jet}}$  is reported in Figure 13. The ratio of the yields is independent of collision energy within errors. For the same kinematic selections, the data at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  correspond to a lower jet energy implying the decrease of the ridge yield with the parton energy, as observed for the jet-like yield.

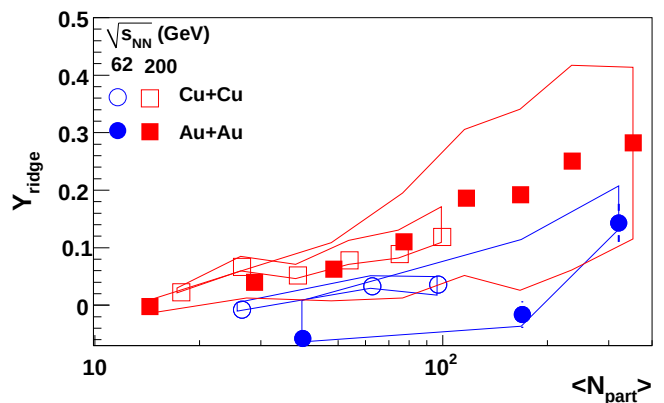


FIG. 12: Dependence of ridge on  $N_{\text{part}}$  for  $3 < p_T^{\text{trigger}} < 6 \text{ GeV}/c$  and  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  for Cu+Cu, and Au+Au at  $\sqrt{s_{\text{NN}}} = 62.4 \text{ GeV}$  and d+Au, Cu+Cu and Au+Au at  $\sqrt{s_{\text{NN}}} = 200 \text{ GeV}$ . Systematic errors due to  $v_2$  are shown as solid lines. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown. Color online.



| System            | $\sigma/\sigma_{\text{geo}}$<br>[%] | $N_{\text{part}}$ | $N_{\text{coll}}$ | $b_{\Delta\phi}$ | $\langle v_2^{\text{trigger}} \rangle$<br>[%] | $\langle v_2^{\text{assoc}} \rangle$<br>[%] | $\tilde{v}_2^2$<br>[% <sup>2</sup> ] | $\tilde{v}_3^2$<br>[% <sup>2</sup> ] |
|-------------------|-------------------------------------|-------------------|-------------------|------------------|-----------------------------------------------|---------------------------------------------|--------------------------------------|--------------------------------------|
| d+Au<br>200 GeV   | MB                                  | 8.3±0.4           | 7.5±0.4           | 0.158±0.006      | 0.0                                           | 0.0                                         | 69 ± 1                               | -285 ± 14                            |
| Cu+Cu<br>62.4 GeV | 0-10                                | 96±3              | 162±14            | 0.885±0.010      | 8.5 <sup>+1.4</sup> <sub>-2.5</sub>           | 7.3 <sup>+1.9</sup> <sub>-0.7</sub>         | 84 ± 5                               | 54 ± 5                               |
|                   | 10-30                               | 64±1              | 87.9±7.9          | 0.515±0.006      | 11.6 <sup>+0.5</sup> <sub>-3.2</sub>          | 10.4 <sup>+0.3</sup> <sub>-1.8</sub>        | 228 ± 5                              | 5 ± 5                                |
|                   | 30-60                               | 25.7±0.6          | 27.6±1.6          | 0.201±0.005      | 14.5 <sup>+1.0</sup> <sub>-5.3</sub>          | 11.8 <sup>+0.3</sup> <sub>-2.7</sub>        | 471 ± 10                             | -77 ± 10                             |
| Au+Au<br>62.4 GeV | 0-10                                | 320±5             | 800 ± 74          | 3.582±0.019      | 8.5 <sup>+0.3</sup> <sub>-3.2</sub>           | 7.6 <sup>+0.3</sup> <sub>-1.9</sub>         | 63 ± 3                               | 42 ± 2                               |
|                   | 10-40                               | 169±9             | 345 ± 44          | 1.846±0.010      | 17.5 <sup>+0.3</sup> <sub>-3.2</sub>          | 15.0 <sup>+0.3</sup> <sub>-1.9</sub>        | 258 ± 2                              | 57 ± 2                               |
|                   | 40-80                               | 42±8              | 51 ± 16           | 0.446±0.009      | 21.2 <sup>+0.3</sup> <sub>-5.9</sub>          | 18.6 <sup>+0.3</sup> <sub>-3.4</sub>        | 456 ± 8                              | -33 ± 8                              |
| Cu+Cu<br>200 GeV  | 0-10                                | 99±1              | 189±15            | 1.759±0.007      | 8.0 <sup>+0.7</sup> <sub>-3.0</sub>           | 8.8 <sup>+0.1</sup> <sub>-2.2</sub>         | 128 ± 2                              | 65 ± 1                               |
|                   | 10-20                               | 75±1              | 123.6±9.4         | 1.206±0.006      | 10.5 <sup>+0.6</sup> <sub>-2.5</sub>          | 11.3 <sup>+0.0</sup> <sub>-1.8</sub>        | 212 ± 2                              | 62 ± 2                               |
|                   | 20-30                               | 54±1              | 77.6±5.4          | 0.815±0.006      | 11.2 <sup>+0.7</sup> <sub>-2.9</sub>          | 12.5 <sup>+0.1</sup> <sub>-1.9</sub>        | 296 ± 3                              | 47 ± 3                               |
|                   | 30-40                               | 38±1              | 47.7±2.8          | 0.540±0.006      | 10.7 <sup>+0.9</sup> <sub>-2.6</sub>          | 13.0 <sup>+0.1</sup> <sub>-2.2</sub>        | 380 ± 4                              | 9 ± 4                                |
|                   | 40-50                               | 26.2±0.5          | 29.2±1.6          | 0.337±0.005      | 11.2 <sup>+2.2</sup> <sub>-6.3</sub>          | 12.6 <sup>+0.5</sup> <sub>-3.7</sub>        | 506 ± 6                              | -37 ± 6                              |
|                   | 50-60                               | 17.2±0.4          | 16.8±0.9          | 0.209±0.005      | 10.3 <sup>+0.6</sup> <sub>-2.7</sub>          | 12.3 <sup>+0.0</sup> <sub>-2.6</sub>        | 657 ± 10                             | -162 ± 10                            |
| Au+Au<br>200 GeV  | 0-12                                | 316±6             | 900±71            | 13.255±0.003     | 8.5 <sup>+2.1</sup> <sub>-2.1</sub>           | 7.3 <sup>+1.5</sup> <sub>-1.5</sub>         | 81 ± 1                               | 49.9 ± 0.2                           |
|                   | 10-20                               | 229±5             | 511±34            | 4.683±0.007      | 14.6 <sup>+2.0</sup> <sub>-2.0</sub>          | 13.0 <sup>+1.2</sup> <sub>-1.2</sub>        | 225 ± 1                              | 67.1 ± 0.6                           |
|                   | 20-30                               | 164±5             | 325±23            | 3.222±0.006      | 17.8 <sup>+2.1</sup> <sub>-2.1</sub>          | 16.5 <sup>+1.2</sup> <sub>-1.2</sub>        | 344 ± 1                              | 71.2 ± 0.8                           |
|                   | 30-40                               | 114±5             | 199±16            | 2.094±0.006      | 18.9 <sup>+2.3</sup> <sub>-2.3</sub>          | 18.1 <sup>+1.4</sup> <sub>-1.4</sub>        | 430 ± 1                              | 68 ± 1                               |
|                   | 40-50                               | 76±5              | 115±12            | 1.267±0.006      | 11.5 <sup>+2.3</sup> <sub>-2.3</sub>          | 19.0 <sup>+2.5</sup> <sub>-2.5</sub>        | 461 ± 2                              | 57 ± 2                               |
|                   | 50-60                               | 48±5              | 61±8              | 0.738±0.006      | 10.1 <sup>+2.3</sup> <sub>-2.3</sub>          | 17.6 <sup>+2.8</sup> <sub>-2.8</sub>        | 478 ± 3                              | 18 ± 3                               |
|                   | 60-70                               | 28±4              | 30±5              | 0.386±0.006      | 8.5 <sup>+2.2</sup> <sub>-2.2</sub>           | 15.4 <sup>+2.9</sup> <sub>-2.9</sub>        | 549 ± 6                              | -25 ± 6                              |
|                   | 70-80                               | 15±2              | 14±3              | 0.183±0.006      | 6.6 <sup>+1.9</sup> <sub>-1.9</sub>           | 12.8 <sup>+2.8</sup> <sub>-2.8</sub>        | 754 ± 12                             | -195 ± 12                            |

TABLE III: The background terms  $b_{\Delta\phi}$  and elliptic flow values of trigger ( $v_2^{\text{trigger}}$ ) and associated particles ( $v_2^{\text{assoc}}$ ) for different collision energies and systems for different centrality bins defined by the fraction of geometric cross section ( $\sigma/\sigma_{\text{geo}}$ ), number of participants ( $N_{\text{part}}$ ) and binary collisions ( $N_{\text{coll}}$ ) for the data in Figure 8.

A recent STAR study of the ridge using two-particle azimuthal correlations with respect to the event plane [14] in Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV shows that the ridge yield is dependent on the angle of the trigger particle relative to the reaction plane. Another STAR study of three-particle correlations in pseudorapidity [13] in Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV shows that within current experimental precision particles in the ridge are uncorrelated in  $\Delta\eta$  with both the trigger particle and each other. These observations along with the PHOBOS measurement of the ridge out to  $\Delta\eta = 4$  [15] provide substantial experimental constraints on theories for the production of the ridge. In addition, CMS recently observed the ridge in high multiplicity p+p collisions at  $\sqrt{s} = 7$  TeV [59]. We consider these results in addition to the results presented in this paper in order to evaluate our current theoretical understanding of the ridge.

The momentum kick model [18] with the same kinematic selection criteria applied to charged particles describes the increase of the ridge yield with centrality quantitatively in Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV. For a given collision energy the ridge yield for Cu+Cu collisions is predicted to approximately follow that for Au+Au collisions for the same number of participants, however the prediction for Cu+Cu collisions is systematically above that for Au+Au collisions at both energies. This difference between the two systems is not corroborated by the data. For the same nucleus-nucleus collisions at different energies, the ridge yield in the momentum kick model is predicted to scale approximately with the number of medium partons produced per participant which increases with increasing collision energy as  $(\ln\sqrt{s})^2$  [60]. According to this prediction,  $Y_{\text{ridge}}$  should increase by a factor of 1.6 in  $\sqrt{s_{\text{NN}}} = 200$  GeV rela-

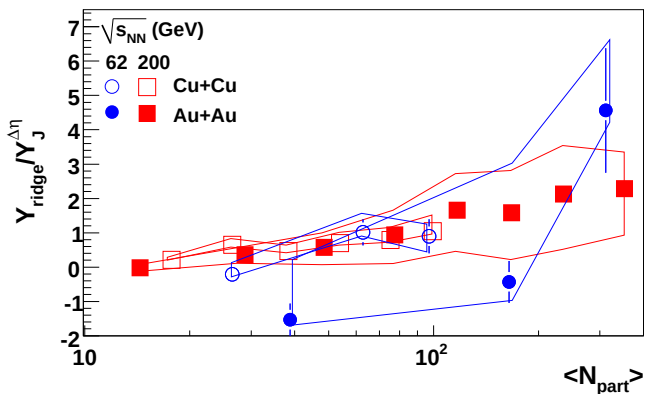


FIG. 13: Ratio of the ridge and jet-like yields as a function of  $N_{\text{part}}$ . Systematic errors due to  $v_2$  are shown as solid lines. Color online.

tive to 62 GeV. This is in agreement with our measurement. We note that the momentum kick model is likely to have difficulty describing the observed dependence of the ridge yield on the event plane, since it would likely predict larger ridge out-of-plane. It would also likely have difficulty explaining the absence of correlations between particles in the ridge and the jet-like correlation.

The model where the ridge arises from the coupling of induced gluon radiation to the longitudinal flow [17] is in qualitative agreement with the observed increase of  $Y_{\text{ridge}}$  as a function of  $N_{\text{part}}$  since the size of the ridge should depend roughly on the average path length traveled by a hard parton. However, it is not obvious that this model can describe the collision system and energy dependence of ridge yield reported in this paper. Moreover, the large extent of the ridge in  $\Delta\eta$  and absence of correlation among ridge particles clearly disfavors this physics mechanism for ridge formation.

The radial flow plus trigger bias model [25, 61, 62] would predict an increasing ridge yield with increasing  $N_{\text{part}}$ . This model would predict a larger ridge in-plane because of the larger surface area in-plane, in agreement with the data. Since the ridge in this model arises from medium partons, particles in the ridge are not expected to be correlated with each other, again in agreement with the 3-particle  $\Delta\eta$  correlation data.

The mechanisms for the production of the ridge through glasma initial state effects [27–29] are able to explain the observed large  $\Delta\eta$  extent of the ridge. However, it is not obvious what this class of models would predict for the collision system and energy dependence discussed in this paper as well as for other ridge properties including its  $p_T^{\text{trigger}}$  and  $p_T^{\text{associated}}$  dependence, the reaction plane dependence, and the absence of correlated structures in three-particle  $\Delta\eta$  correlations. While there are calculations of untriggered di-hadron correlations for the glasma model, there are no calculations to compare to high- $p_T$  triggered correlations. It should be noted that these calculations not only include the glasma initial state but also hydrodynamical flow so some of the

ridge in these models is created by flow. If the ridge is produced by the same mechanism in p+p and A+A collisions, models where the ridge is produced by initial state effects such as the glasma model may be the only models able to explain both the p+p and A+A data simultaneously since hydrodynamical flow in p+p collisions is expected to be small if not negligible. Therefore quantitative calculations of the ridge in this class of models and the identified particle spectra measurements in high multiplicity p+p collisions are essential for understanding of the production mechanism of the ridge.

Models describing the ridge in terms of  $v_n$  from initial eccentricity fluctuations predict a ridge yield that increases with  $N_{\text{part}}$  [31, 33, 66] in qualitative agreement with data. Motivated by these models, we applied two dimensional fits to our data with a two dimensional Gaussian to describe the jet-like component and  $\Delta\eta$  independent  $\tilde{v}_n^2$  terms given by Eq. (11). Figure 14 shows  $(\tilde{v}_3/\tilde{v}_2)^2$  as a function of  $N_{\text{part}}$  from these fits. The values of the Fourier coefficients  $\tilde{v}_3^2$  and  $\tilde{v}_2^2$  are given in Table III. We allow  $\tilde{v}_3^2$  to be negative and  $\tilde{v}_3^2$  is negative for d+Au collisions and peripheral A+A collisions. An approximately Gaussian peak from an away-side jet-like correlation would give a negative  $\tilde{v}_3^2$  and indicate that  $\tilde{v}_3^2$  is dominantly from non-flow. The ratio  $(\tilde{v}_3/\tilde{v}_2)^2$  evolves from negative values in d+Au and peripheral A+A collisions to positive values at larger  $N_{\text{part}}$ . Positive values for  $\tilde{v}_3^2$  are consistent with expectations from triangular initial conditions. Contributions to a jet-like peak on the away-side, such as that observed in d+Au, would lead to  $(\tilde{v}_3/\tilde{v}_2)^2 < (v_3/v_2)^2$  because  $\tilde{v}_3^2$  would be an underestimate of  $v_3^2$  and  $\tilde{v}_2^2$  would be an overestimate of  $v_2^2$ . For both Au+Au and Cu+Cu collisions, the ratio is independent of collision energy within errors and is largest in the most central collisions with significant deviation between the two colliding systems at the same  $N_{\text{part}}$ . It is not clear at this point whether the representation of the data in Figure 12, where the  $v_2$  has been subtracted, or in Figure 14, where the  $v_2$  is explicitly included, gives the most insight into the production mechanism of the ridge.

We compare the data to three hydrodynamical models in Figure 14. The theory calculations for Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV from [63] in Figure 14a, a hydrodynamical model with a lattice equation of state and a viscosity to entropy ratio  $\eta/s = 1/4\pi$ , agree with the data for all but the three most peripheral centrality bins for a freeze-out temperature of 170 MeV. Comparisons to [64] were only possible for Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV with momenta of  $1.5 < p_T < 4.0$  GeV/c for both  $p_T^{\text{trigger}}$  and  $p_T^{\text{associated}}$ . Figure 14b shows this model also reproduces the trend observed in the data. For comparisons with the 3+1D ideal hydrodynamic calculation embedded into the transport model UrQMD [65] in Figure 14c, we scaled the theory by 0.5. Requiring a high  $p_T$  trigger particle skews the calculation towards events with a hot spot in the initial density profile, which may lead to a preference for events with a high  $v_3$  value. This model accurately describes the  $N_{\text{part}}$  dependence of  $(\tilde{v}_3/\tilde{v}_2)^2$  for

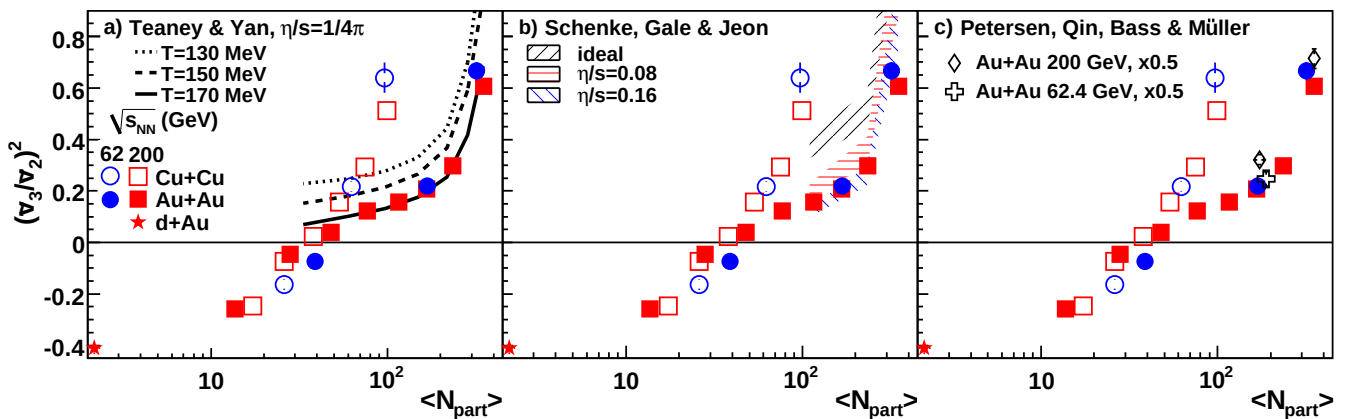


FIG. 14:  $(\tilde{v}_3/\tilde{v}_2)^2$  ratio as a function of  $N_{\text{part}}$ . Statistical errors only. The data are compared to hydrodynamical calculations using (a) lattice equation of state with three different freeze-out temperatures in Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV from [63] (b) using a 3+1D viscous hydrodynamical model with three different viscosity to entropy ratios,  $\eta/s$  [64], and (c) a 3+1D model incorporating both hydrodynamics and transport model [65]. Color online.

Au+Au collisions at both  $\sqrt{s_{\text{NN}}} = 62.4$  GeV and 200 GeV. Hydrodynamical models are able to describe the  $N_{\text{part}}$  dependence of  $(\tilde{v}_3/\tilde{v}_2)^2$ . However, if there are substantial non-flow contributions the data may favor different values of the model parameters. Additionally, this kinematic region is difficult for hydrodynamical models and viscous corrections are large.

If the description of the ridge in terms of initial eccentricity fluctuations is correct, then these dihadron correlations effectively represent an image of tube-like structures in the initial conditions of heavy-ion collisions [27, 67]. However, detailed comparisons of these models' predictions for other observables, such as particle spectra and  $v_2$ , to data are necessary to assess the validity of the theories. In addition, future theoretical and experimental studies will be needed to determine whether these models are sufficient to describe the ridge and whether there are substantial non-flow contributions to the Fourier coefficient  $\tilde{v}_3^2$  in central A+A collisions.

## V. CONCLUSIONS

The energy and system size dependence of near-side di-hadron correlations enables studies of the jet-like correlation and the ridge at fixed densities with different geometry. There is remarkably little dependence of the jet-like correlation on the collision system at both  $\sqrt{s_{\text{NN}}} = 62.4$  GeV and 200 GeV except for central Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV. In central Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV the jet-like correlation is substantially broader and the spectrum softer than in peripheral collisions and than those in collisions of other systems in this kinematic regime. This may indicate that fragmentation is modified in these collisions so that the parton fragments softer, perhaps due to a mechanism such as gluon bremsstrahlung. The reasonable agreement with PYTHIA is surprising, especially considering that

PYTHIA is a p+p event generator. This indicates that the near-side jet-like correlation is dominantly produced by vacuum fragmentation.

The ridge is observed not only in Au+Au collisions at  $\sqrt{s_{\text{NN}}} = 200$  GeV as we reported earlier, but also in Cu+Cu collisions and in both studied collision systems at lower energy of  $\sqrt{s_{\text{NN}}} = 62.4$  GeV. This demonstrates that the ridge is not a feature unique to Au+Au collisions at the top RHIC energy. We observe two trends which set significant limits to models. First, when the ridge is measured using the standard ZYAM model, the ridge is comparable in Cu+Cu and Au+Au collisions and the energy dependence of the ridge is the same as the energy dependence of the jet-like correlation. Second, when we subtract the jet-like correlation and calculate the third component of the Fourier decomposition  $\tilde{v}_3$ , which is  $v_3$  in the absence of non-flow, we see different trends for Cu+Cu and Au+Au but no difference between the two energies. The combination of these data with future measurements at lower RHIC energies ( $\sqrt{s_{\text{NN}}} = 7$ -39 GeV) as well as studies at the LHC will therefore be a powerful tool for the distinction between various theoretical models for the production of the ridge.

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